

Vatsal Thakkar

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SUMMARY:

Cloud Infrastructure & AI Platform Engineer with 3+ years of experience designing, developing and deploying ML/LLM systems at scale. Proven expertise in IaC, K8s, GitOps, and Azure/AWS/GCP, delivering secure, highly available, and cost-optimized solutions, and AI/ML workloads. Strong background in MCP, RAG, LLM fine-tuning (LoRA), and inference optimization using NVIDIA NIM, vLLM and Triton.

SKILLS:

Programming Languages: Python, JavaScript, Java, C++, C, R, TypeScript, Linux Shell Scripting, HTML, CSS
Python Libraries: PyTorch, Transformers, vLLM, Numba(CUDA), LangChain, Pandas, NumPy, Scikit-Learn, LlamaIndex, WandB
Frameworks: FastAPI, Flask, Django, React.js, Node.js, Spring, Bootstrap, .NET
Database Management: MySQL, Oracle, SQL, MongoDB (NoSQL), PostgreSQL, RDS, DynamoDB, Qdrant, Redis
Cloud Services: AWS (EC2, Lambda, SageMaker, S3, CloudWatch, ECS, ECR, Route53, VPC), GCP (Vertex AI), Azure, Azure DevOps
Other Tools and Skills: Git, GitHub, Docker, Airflow, Kafka, Kubernetes, MLflow, Terraform, NLP, LLMs, Generative AI (GenAI), CI/CD

PROFESSIONAL EXPERIENCE:

Cloud Infrastructure Engineer | ASML Holding, United States

Dec 2025 – Present

- Architected and delivered enterprise-scale Managed Infrastructure Platforms enabling secure self-service onboarding for compute, database, storage, IaaS, PaaS, and AI/ML workloads, **reducing environment provisioning time by ~60-70%**.
- Designed and scaled a modular Terraform-based IaC architecture, standardizing provisioning for AKS, VMs, Databricks, Data Lakes, Azure OpenAI, managed databases, storage, and GPU workloads, reducing **configuration drift by ~80%**.
- Standardized **Kubernetes** operations across the organization by implementing **GitOps** workflows and automated security policies, achieving **99.99% system availability** and zero-downtime deployments for critical microservices.

AI Engineer-DevOps/MLOps | ASML Holding, United States

Nov 2024 – Nov 2025

- Led large-scale migration of internal DevOps tools (**Bitbucket, Bamboo, SonarQube, Bamboo Agents, Ray**) using Terraform, K8s, and Helm, **reducing developer cycle time by 70%** through automated environment provisioning and CI/CD pipeline optimization.
- Built an internal **Confluence GPT** agent using **RAG pipelines** and **LLMs** to improve documentation discovery.
- Deployed **latency-optimized LLM inference** workloads on AKS, using GPU-accelerated **NVIDIA NIM** containers for LLM applications.
- Developed **organization-wide IaC standardization**, designing scalable, modular **Terraform** frameworks to enable complete infrastructure provisioning, GPU cluster setup, and future-ready application deployments across Kubernetes environments.

LLM and AI Engineer | ColomboAI, United States

Dec 2023 – Nov 2024

- Architected and developed a low-latency **RAG (Retrieval Augmented Generation) pipeline** with a **Mixture-of-Experts** strategy for Search Agent LM, enabling dynamic model selection based on user queries, mitigating hallucination in **LLM-generated** responses.
- Deployed and monitored **throughput/latency optimized** Generative models (**Diffusion, LLM, VLMs**) on **GCP Vertex AI** utilizing **vLLM** and **Triton inference server** endpoints.
- Fine-tuned LLMs** using **LoRA** on **NVIDIA** accelerator to enhance a search retrieval pipeline, boosting relevance accuracy by 12%.
- Developed **ETL Pipeline** for LLM-driven audio summarizer as an event-driven serverless workflow using **AWS Lambda and Bedrock**.

ML Research Associate | The Hoarfrost Lab, University of Georgia, United States

Aug 2023 – Jun 2024

- Designed and implemented a **multimodal language model** that integrates textual and DNA sequence data, enhancing the model's ability to interpret and generate biological insights.
- Developed an **ETL Pipeline** to process extensive DNA sequence and textual description data, utilizing **Apache Airflow**.

Junior Software Engineer | ECPL Edtech, India

Jan 2021 – Jul 2022

- Developed a hybrid course recommendation system with collaborative filtering, content-based algorithms, and RNNs to deliver personalized course suggestions, increasing user engagement by **10%**.
- Built and trained an **RNN** model to predict student course completion likelihood based on sequential learning behaviour, enabling targeted interventions and achieving a **15% improvement in overall course completion** rates.
- Created an intuitive and actionable web interface by leveraging data visualization techniques and **implementing RESTful APIs**, enabling users to gain valuable insights for informed decision-making, resulting in a significant **25% reduction in data analysis time**.

EDUCATION:

Master of Science in Computer Science | The University of Georgia, United States

Aug 2022 – May 2024

Grade (GPA): 3.89 / 4.0

Coursework: Computer Vision and Deep Learning, Representation Learning, Data Science, Distributed Systems, Database Management.

Bachelor of Engineering in Computer Engineering | L. D. College of Engineering, India

Aug 2017 – Jul 2021

Grade (GPA): 3.93 / 4.0

Coursework: Data Structures and Algorithms, Software Engineering, Object-oriented Programming, Artificial Intelligence, Operating System.

PROJECT EXPERIENCE:

Decoding the Absence: Negative Prompts in Stable Diffusion | Diffusers, PyTorch, Transformers, Torchvision

Aug 2024 – Jan 2024